

# Genetic Programming Hyper-Heuristics for the Job Shop Scheduling Problem with Interval Durations: Interpretable Rules that Generalize Across Instance Sizes

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## Abstract

Dispatching rules remain the method of choice for scheduling under tight time budgets, but hand-crafted rules leave substantial performance unexploited, and the automated design of rules has scarcely been explored for scheduling problems with non-deterministic processing times. We present a genetic programming (GP) hyper-heuristic for the job shop scheduling problem with interval durations (IJSP), in which operation durations are known only to lie within an interval and schedules are evaluated by interval arithmetic. The GP evolves priority expressions over an *interval-aware* terminal set that exposes both the worst-case values and the widths of the uncertain quantities, so that reacting to uncertainty itself is expressible. On the 70 interval versions of Taillard’s benchmark, over 30 independent evolutions the resulting rules average a relative error of 18.7% ( $\pm 0.8$ ) with a single deterministic constructive pass, and the best evolved rule attains 17.3%—improving on the strongest deterministic constructive baseline (Giffler–Thompson with MWKR tie-breaking, 29.5%) on every single instance—and generalizes zero-shot across all seven instance size classes despite being evolved on only four

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training instances of a single size class ( $20 \times 15$ ). Embedded in a GRASP-style randomized scheme, the evolved rule converts sampling budget into solution quality (13.7% at best-of-1024). Evolved rules are compact, human-readable expressions, and an automatic configuration study with irace shows the approach is robust to its own hyperparameters. A controlled ablation over 30 evolutions per arm characterizes *when* interval awareness pays: the width terminals do not improve the expected makespan—the worst-case bounds carry that signal—but under an objective that rewards predictability they yield significantly narrower makespan intervals, a trade-off the user can dial. We position the method against fixed rules, active schedule generators, and published metaheuristics, characterizing the computational class in which interpretable evolved rules operate.

*Keywords:* interval job shop scheduling, genetic programming, hyper-heuristics, dispatching rules, scheduling under uncertainty, robustness

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## 1. Introduction

Real manufacturing environments rarely know the exact duration of an operation in advance. The interval job shop scheduling problem (IJSP) models this uncertainty in a minimal, distribution-free way: each processing time is only assumed to lie within a known interval, and schedules are built and evaluated with interval arithmetic, so that the makespan of a solution is itself an interval and solutions are compared by an interval ranking. Population metaheuristics for the IJSP—artificial bee colony variants and tabu search among them—achieve strong results, at the cost of building and evaluating large numbers of complete solutions for every instance [2, 3].

At the opposite end of the computational spectrum live *dispatching rules*:

priority functions that build a schedule in a single constructive pass by repeatedly selecting the next operation to start. Their appeal is threefold—speed, interpretability, and trivial deployment—but hand-crafted rules (SPT, MOR, MWKR and relatives) leave a large quality gap with respect to search-based methods, as the experiments reported later in this paper confirm.

For *deterministic* scheduling, a rich literature has shown that this gap can be narrowed substantially by evolving rules automatically with genetic programming (GP) [18, 19]. For scheduling under interval uncertainty, however, the automated design of dispatching rules remains, to the best of our knowledge, unexplored. Moving automated rule design to interval durations raises questions that simply do not arise in the deterministic case: which attributes a rule should read, when the quantities describing a candidate operation—its duration, its earliest start, the work remaining in its job—are intervals rather than numbers; how candidate operations should be prioritized, and the resulting schedules compared, given that intervals admit no natural total order; and whether the *magnitude* of the uncertainty carries information that is useful for dispatching, or it is the position of the intervals that drives the decision. Answering these questions empirically requires exposing uncertainty to the rule as first-class attributes, which is the starting point of this work.

This paper makes the following contributions:

1. A GP hyper-heuristic for the IJSP whose terminal set is *interval-aware*: alongside worst-case processing time, earliest start and work remaining, rules can read the *widths* of these quantities, so that uncertainty-sensitive prioritization is expressible at all (Section 4).
2. An empirical study on the 70 interval Taillard instances showing that

evolved rules (i) beat every deterministic constructive baseline by a wide margin at equal cost, (ii) generalize zero-shot across instance sizes—evolved on  $20 \times 15$ , best class  $50 \times 15$ —and (iii) extend naturally to a randomized multi-start regime with monotone quality gains up to best-of-1024 (Section 6).

3. An analysis of *why* the rules work: an anatomy of the evolved expressions (Section 7.1); an ablation that isolates the contribution of the interval-width terminals and characterises *when* interval awareness pays—not under a makespan objective, where worst-case bounds suffice, but under an objective that rewards predictability, where the widths yield significantly narrower makespan intervals (Section 7.2); an executional-robustness study showing that the evolved rule yields schedules whose realised makespan deviates less from its prediction than the baselines’, increasingly so as uncertainty grows (Section 7.3); and a transfer study showing that the resulting schedules remain competitive when re-evaluated under crisp and fuzzy uncertainty models (Section 7.4).
4. An automatic configuration study (irace, 200 experiments) of the evolutionary hyperparameters, confirmed over 30 independent evolutions per configuration: tuning brings no significant quality gain—the method is robust to its own knobs—but does yield systematically smaller, more readable trees at equal performance (Section 5.3).

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 formalizes the IJSP. Section 4 presents the GP hyper-heuristic, from both the hyper-heuristic and the evolutionary viewpoints. Section 5 details the benchmark, the training protocol and the irace

configuration study. Section 6 reports the empirical results, and Section 7 analyses them: rule anatomy, the interval-awareness ablation, executional robustness, transfer to crisp and fuzzy uncertainty models, and limitations. Section 8 concludes.

## 2. Related Work

### *2.1. Job shop scheduling under interval uncertainty*

Interval durations offer a distribution-free uncertainty model requiring only bounds, in contrast to stochastic and fuzzy approaches [27]: they are the natural choice when historical data are insufficient to estimate distributions and a decision maker can only supply a minimum and a maximum duration. Scheduling with interval processing times has been studied across machine environments—single machine [12], flow shop, and flexible job shop [14], among others surveyed in [13]—and also analytically, through measures of the problem uncertainty induced by the intervals and of the stability of optimal sequences [11].

For the job shop with makespan minimisation, which concerns us here, the literature is dominated by population-based metaheuristics. Lei’s population-based neighbourhood search [10], which ranks interval makespans by a possibility degree, was an early reference point; it was subsequently outperformed by a genetic algorithm with a systematic study of interval rankings [1] and by artificial bee colony variants [2, 3], which report the best published results for this problem. Related objectives have also been addressed under interval uncertainty, notably tardiness minimisation [15]. All of these are iterative search methods: they build and evaluate a population of complete solutions and refine it over many

iterations for each instance solved. Fast constructive methods for the IJSP have received far less attention; dispatching rules with interval-aware lexicographic comparisons are the de facto baseline.

## 2.2. Dispatching rules for job shop scheduling

Priority dispatching rules are among the oldest and most widely deployed scheduling heuristics: at every decision point, a priority function ranks the operations that are ready to start and dispatches the best one, building a schedule in a single constructive pass. Classical rules score each ready operation by a single attribute, and we adopt their standard acronyms throughout this paper. *Shortest* and *longest processing time* (SPT, LPT) look at the duration of the operation itself. *Earliest starting time* (EST) prefers the operation that can begin soonest given the current state of its job and its machine. *Most work remaining* (MWKR) and *most operations remaining* (MOR) look instead at the job the operation belongs to, and favour the one with the largest amount of pending work, measured respectively in time units and in number of operations. The *critical ratio* (CR) weighs the work still to be done against the time available until a due date. Surveys catalogue more than a hundred such variants [7, 8]. A step above single-attribute rules, the algorithm of Giffler and Thompson (G&T) [4] restricts each choice to a machine’s conflict set, guaranteeing *active* schedules and markedly improving plain rules when a priority criterion is used to break ties inside that set; we write G&T-SPT and G&T-MWKR for the resulting combinations. Stronger constructive procedures exist for the *deterministic* job shop, notably the shifting bottleneck heuristic [5], which repeatedly solves one-machine subproblems with heads and tails by means of Carlier’s algorithm [6]. These fall outside the scope of the present work: their machinery rests on exact

one-machine optimisation and critical-path arguments whose extension to interval arithmetic, where no natural total order is available, is a research contribution in itself and remains open. Decades of practice consolidated two lessons: rules are unbeatable in speed, transparency and ease of deployment, but hand-crafting them plateaued—no fixed combination of attributes performs well across shop configurations and objectives [8]. That plateau is precisely what motivated the *automated* design of rules.

### 2.3. Automated design of dispatching rules

Genetic programming [16, 17] evolves a population of expressions, usually represented as trees, by applying selection and variation operators directly on their structure. GP hyper-heuristics apply this machinery to *priority expressions* over scheduling attributes, and constitute the standard approach to automated rule design for production scheduling [18, 19]. Evolved rules have been shown to outperform hand-crafted ones across job shop variants, with active research on surrogate fitness, feature selection, and multi-objective formulations. Beyond single rules, a productive line combines several evolved rules into *ensembles* that jointly construct or vote on a solution [22], and recent work explores alternatives to evolutionary search for the same design task—systematic tree search over the space of expression trees [23]—reporting rules of comparable quality but smaller size and better readability. All of this work targets deterministic (or dynamic-stochastic) settings. Closest to our setting, Karunakaran et al. [20] evolve rules for dynamic job shops with uncertain processing times, exposing summary statistics of observed deviations (an exponential moving average) to the rule; uncertainty there is realized stochastically during simulation, and rules are evaluated on sampled scenarios. A recent survey of learned optimisation for the job shop [21],

covering both GP and DRL constructive approaches, records no work on interval-valued processing times. To the best of our knowledge, evolving rules whose *inputs* are interval attributes—and whose fitness is computed by interval arithmetic rather than by sampling realizations—has not been explored before.

Finally, deep reinforcement learning provides an alternative family of learned constructive methods for the JSP [25, 26]; GP rules and DRL policies are increasingly studied as competing learning paradigms for the same construction task, and controlled comparisons between them under common protocols have been identified as a gap in the recent literature [21]. Both families occupy the same computational class at deployment: a single constructive pass.

### 3. The Interval Job Shop Scheduling Problem

An IJSP instance consists of  $n$  jobs and  $m$  machines. Job  $j$  is a chain of  $m$  operations  $o_{j1} \prec o_{j2} \prec \dots \prec o_{jm}$ , each requiring a distinct machine  $\mu(o_{jk})$  exclusively and non-preemptively. The duration of operation  $o$  is an interval  $p_o = [\underline{p}_o, \bar{p}_o]$  with  $0 < \underline{p}_o \leq \bar{p}_o$ : the only assumption is that the realized duration lies within these bounds.

Schedule construction uses interval arithmetic. For intervals  $a = [\underline{a}, \bar{a}]$  and  $b = [\underline{b}, \bar{b}]$ , the two operations required by makespan minimisation are

$$a + b = [\underline{a} + \underline{b}, \bar{a} + \bar{b}], \tag{1}$$

$$\max(a, b) = [\max(\underline{a}, \underline{b}), \max(\bar{a}, \bar{b})], \tag{2}$$

both component-wise [1]. A solution determines a task processing order  $\pi$  (a permutation with repetition of jobs, decoded left to right). Following the



notation of [1, 3], let  $PM_o(\pi)$  and  $PJ_o$  denote the predecessors of task  $o$  in its machine (according to  $\pi$ ) and in its job. The *semi-active* schedule [9] starts each task at

$$\mathbf{s}_o(\pi) = \max(\mathbf{c}_{PJ_o}(\pi), \mathbf{c}_{PM_o(\pi)}(\pi)), \quad \mathbf{c}_o(\pi) = \mathbf{s}_o(\pi) + \mathbf{p}_o, \quad (3)$$

so starting and completion times of all tasks are themselves intervals. The problem can be stated as

$$\min_R \mathbf{C}_{\max} \quad (4)$$

$$\text{s.t. } \mathbf{C}_{\max} = \max_{1 \leq j \leq n} \{\mathbf{c}_{o(j, m_j)}\}, \quad (5)$$

$$\underline{s}_{o(j, l)} \geq \underline{c}_{o(j, l-1)}, \quad \bar{s}_{o(j, l)} \geq \bar{c}_{o(j, l-1)}, \quad 1 \leq l \leq m_j, \quad 1 \leq j \leq n, \quad (6)$$

$$\underline{s}_o \geq \underline{c}_{o'} \vee \underline{s}_{o'} \geq \underline{c}_o, \quad \bar{s}_o \geq \bar{c}_{o'} \vee \bar{s}_{o'} \geq \bar{c}_o, \quad \forall o \neq o' : \mu(o) = \mu(o'), \quad (7)$$

where Eq. (5) defines the interval makespan as the component-wise maximum of the job completion times, Eq. (6) enforces precedence within each job and Eq. (7) forbids overlaps on each machine, in both interval components [1, 3]. The minimum in (4) is taken with respect to an interval ranking  $R$ , since intervals admit no natural total order. We rank schedules lexicographically, comparing upper bounds first and using the lower bound only to break ties:

$$\mathbf{a} \preceq \mathbf{b} \iff \bar{a} < \bar{b} \vee (\bar{a} = \bar{b} \wedge \underline{a} \leq \underline{b}), \quad (8)$$

one of the standard rankings of the IJSP literature and the one found in [1] to yield the most robust schedules. All methods in this paper—the GP rules and every baseline—share one implementation of this decoder and evaluator, so reported differences cannot stem from evaluator discrepancies.

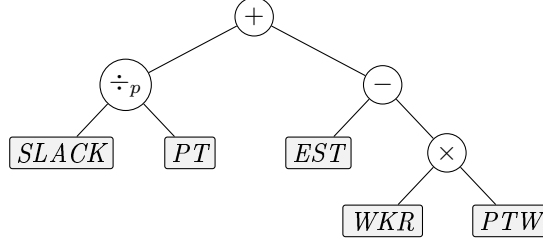


Figure 1: An individual is a priority expression tree; leaves are interval-aware attributes of the candidate operation (here the illustrative rule  $SLACK/PT + EST - WKR \cdot PTW$ ). At each decision point the tree is evaluated on every eligible operation and the lowest-priority one is dispatched.

Performance is reported as the relative error of the expected makespan with respect to a crisp reference bound  $LB$  of each instance,

$$RE = \frac{E[\mathbf{C}_{\max}] - LB}{LB} \times 100, \quad E[\mathbf{C}_{\max}] = \frac{\underline{C}_{\max} + \overline{C}_{\max}}{2}, \quad (9)$$

where  $E[\mathbf{C}_{\max}]$  is the expected value of the interval makespan under the uniform distribution on  $[\underline{C}_{\max}, \overline{C}_{\max}]$ , and  $LB$  is a bound of the underlying deterministic instance—Taillard’s bounds for the Taillard-derived instances, the best known ones [2] for the classical set. It serves as an indicative common reference, not as a bound for the interval problem, for which none is available.

#### 4. GP Hyper-Heuristic for the IJSP

We describe the method from two complementary viewpoints: as a *hyper-heuristic*—what an individual is and how it turns an IJSP instance into a schedule (Section 4.1)—and as an *evolutionary algorithm*—how the population of rules is initialized, varied and selected (Section 4.2).

Table 1: Interval-aware terminal set, with exact definitions. For an eligible operation  $o$  of job  $j$  on machine  $m$ :  $p_o = [\underline{p}_o, \bar{p}_o]$  is its duration interval;  $C_j$  and  $C_m$  are the (interval) completion times of the last scheduled operation of job  $j$  and machine  $m$ ;  $\mathcal{R}_o$  is the set of operations of  $j$  after  $o$ ;  $\mathcal{E}$  is the current eligible set. Terminals are evaluated on *raw* (unnormalized) values.

| Terminal | Definition                                                                                      | Role                            |
|----------|-------------------------------------------------------------------------------------------------|---------------------------------|
| PT       | $\bar{p}_o$                                                                                     | worst-case processing time      |
| PTW      | $\bar{p}_o - \underline{p}_o$                                                                   | duration uncertainty            |
| EST      | $\bar{e}_o = \max(\bar{C}_j, \bar{C}_m)$                                                        | worst-case earliest start       |
| ESTW     | $\bar{e}_o - \underline{e}_o$ , with $\underline{e}_o = \max(\underline{C}_j, \underline{C}_m)$ | start uncertainty               |
| WKR      | $\sum_{q \in \mathcal{R}_o} \bar{p}_q$                                                          | worst-case work remaining       |
| WKRW     | $\sum_{q \in \mathcal{R}_o} (\bar{p}_q - \underline{p}_q)$                                      | remaining-work uncertainty      |
| NOR      | $ \mathcal{R}_o $                                                                               | operations remaining            |
| SLACK    | $\bar{e}_o - \min_{o' \in \mathcal{E}} \bar{e}_{o'}$                                            | start slack within eligible set |
| ONE      | 1                                                                                               | constant                        |

#### 4.1. Hyper-heuristic view: from expression trees to schedules

An individual is a priority expression tree (Figure 1). Leaves are drawn from the interval-aware terminal set of Table 1, and internal nodes from the function set

$$\mathcal{F} = \{+, -, \times, \div_p, \min, \max, \text{neg}\}, \quad (10)$$

where  $\text{neg}$  is the unary sign change and  $\div_p$  denotes protected division, defined as  $a \div_p b = a/b$  if  $|b| > 10^{-9}$  and  $a$  otherwise. The rule is deployed inside a semi-active schedule generation scheme (SGS): starting from the empty schedule, at each of the  $N$  decision points the set  $\mathcal{E}$  of eligible operations (those whose job predecessor is already scheduled) is formed, the tree is evaluated on the attributes of every  $o \in \mathcal{E}$ , and the operation with *lowest* priority value is appended at its earliest feasible start on its machine, Eq. (3).

Because the lowest value wins, a criterion that seeks to *maximise* an attribute must be expressed through the unary neg operator. Classic rules are thus points of this search space: shortest processing time is the single-node tree  $PT$ , whereas most work remaining is  $\text{neg}(WKR)$ . We inject both as seed individuals in the initial population, so that the search departs from a population that already contains well-known rules rather than from purely random expressions.

Priority ties at dispatch time are broken by eligible set order (lowest job index first), which is deterministic; hence a rule maps each instance to exactly one schedule.

#### 4.2. Evolutionary components

The initial population is built with the *ramped half-and-half* method, introduced by Koza [16] and still the standard choice for tree-based GP [17], which combines two tree-generation procedures over a range of depths. The *full* method expands every branch with function nodes until the prescribed maximum depth is reached, placing terminals only at that level, and thus produces complete trees whose leaves all lie at the same depth, whereas the *grow* method may close a branch earlier by placing a terminal—with probability 0.3 at each node in our implementation—and thus produces trees of irregular shape and variable size. In both cases the function at an internal node is drawn uniformly from  $\mathcal{F}$  and a leaf uniformly from the terminal set, the arity of the chosen function determining the number of branches to expand. Each individual draws its maximum depth uniformly from  $\{2, 3, 4\}$  and is generated by either method with equal probability. The last two members of the population are not generated at random: they are the two seed rules of Section 4.1, inserted directly. Every individual is evaluated

by the mean RE of its deterministic constructive pass over the four training instances, computed with the same environment, decoder and evaluator used by all baselines, which eliminates evaluator mismatch as a confound.

Each generation then proceeds as follows. Parents are chosen by tournament, drawing  $k$  individuals uniformly without replacement from the current population and returning the fittest of them. With probability  $p_c$  two parents are recombined by *subtree crossover* [16, 17], the standard recombination operator for tree-based GP. The offspring is a copy of the first parent in which one subtree has been replaced by a subtree taken from the second: a node is drawn uniformly at random in each parent, leaves included, and the subtree rooted at the node chosen in the first parent is substituted by the subtree rooted at the node chosen in the second. In its classical formulation the operator exchanges both subtrees and returns two offspring; we keep a single one per recombination. With the complementary probability  $1 - p_c$  a single parent undergoes *subtree mutation* [16, 17]: a node is again drawn uniformly at random, and the subtree rooted at it is discarded and replaced by a freshly generated random tree, built with the grow method and maximum depth 3 over the full primitive set. Offspring whose size exceeds a cap of  $c$  nodes receive infinite fitness and are thereby excluded from selection. Note that the depth bounds above apply only to the trees created from scratch, at initialization and inside mutation: the depth of an individual is never constrained during the run, so recombination routinely produces expressions deeper than those of the initial population, the number of nodes being the only quantity that is bounded.

Replacement is fully generational, the offspring becoming the next population, except for the  $e$  best individuals of the current generation, which are copied into it unchanged. The process stops after a fixed number of

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**Algorithm 1** GP hyper-heuristic for the IJSP

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1:  $P \leftarrow \text{ramped half-and-half}(\text{pop} - 2) \cup \{\text{SPT}, \text{MWKR}\}$ 
2:  $\text{evaluate}(P)$   $\triangleright$  mean RE of one constructive pass on the training
   instances
3: for  $g = 1$  to  $\text{gens}$  do
4:    $P' \leftarrow \text{elite}(P, e)$   $\triangleright$  elitism
5:   while  $|P'| < \text{pop}$  do
6:     if  $\text{rand}() < p_c$  then
7:        $\text{child} \leftarrow \text{subtreeCrossover}(\text{tournament}(P, k),$ 
        $\text{tournament}(P, k))$ 
8:     else
9:        $\text{child} \leftarrow \text{subtreeMutation}(\text{tournament}(P, k))$ 
10:    end if
11:     $P' \leftarrow P' \cup \{\text{child}\}$   $\triangleright$  fitness  $\infty$  if size  $> c$ 
12:  end while
13:   $P \leftarrow \text{evaluate}(P')$ 
14: end for
15: return best rule in  $P$ 
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generations and returns the best rule found. Algorithm 1 summarizes the generational loop and Table 2 lists the parameter values.

#### 4.3. Randomized multi-start extension

A deterministic rule produces one schedule. For matched comparisons with sampling-based methods we wrap the evolved rule in GRASP-style randomization: with probability  $\varepsilon=0.1$  the dispatched operation is drawn uniformly from the eligible set, otherwise the rule decides; sample 0 remains deterministic. Best-of- $N$  then reports the lexicographically best schedule. This

Table 2: Evolution parameters (conventional settings; see Section 5.3 for the sensitivity study).

|                            |                                                |
|----------------------------|------------------------------------------------|
| Population / generations   | 100 / 50                                       |
| Initialization             | ramped half-and-half (depths 2–4) + 2 seeds    |
| Selection                  | tournament, size $k = 4$                       |
| Crossover / mutation prob. | $p_c = 0.8$ / $0.2$ (subtree, depth $\leq 3$ ) |
| Elitism                    | $e = 2$                                        |
| Tree-size cap              | $c = 40$ nodes (violations: fitness $\infty$ ) |
| Fitness                    | mean RE, deterministic rollout, TA11–TA14      |
| Seeds                      | 1–30 (RNG of evolution and environment)        |

uniform perturbation of a single rule is the simplest member of a broader family of randomized rule-based builders; a richer alternative, orthogonal to our contribution, swaps among a *set* of evolved rules during construction [24].

## 5. Experimental Setup

### 5.1. Benchmark instances

We use two benchmarks, both taken from the IJSP literature [1, 2, 3]: the 70 interval versions of Taillard’s instances (TA1–TA70, seven size classes from  $15 \times 15$  to  $50 \times 20$ ), and the 12 classical instances on which the published IJSP metaheuristics report results (FT10, FT20, six instances of the La family and ABZ7–9, of sizes  $10 \times 10$  to  $20 \times 15$ ). Both were obtained from their crisp counterparts with the generation scheme introduced in [1]: for every operation a value  $\delta$  is drawn uniformly at random in  $[0, 0.15p]$ , where  $p$  is its original duration, and that duration becomes the integer interval  $[p - \delta, p + \delta]$ , machine routes being unchanged. The interval is symmetric around  $p$ , so the crisp instance is exactly the midpoint scenario of its interval

version. Since rules are evolved on Taillard instances only and applied to the classical set without any re-evolution, the latter doubles as a cross-family generalization test (Section 6.4). Both sets are released with this paper (see Data availability).

### 5.2. *Training protocol, data split and baselines*

Rules are evolved on TA11–TA14 ( $20 \times 15$ ), and TA15–TA20 serve as development set: they never contribute to fitness, but design and configuration decisions (including the irace study of Section 5.3) were selected on their performance, so we flag them as subject to development reuse. The remaining 60 Taillard instances and the 12 classical instances are fully unseen by every design decision and constitute the honest generalization test.

Each evolution uses population 100 and 50 generations:  $\sim 20,400$  fitness evaluations ( $\approx 30$  min single-core) per seed; thirty independent seeds per configuration, following standard practice in evolutionary computation.

### 5.3. *Hyperparameter configuration with irace*

The four qualitative knobs of the evolution—tournament size, crossover probability, tree-size cap and elitism—were configured with irace over a budget of 200 experiments, each a full evolution (population 100, 50 generations) on the training instances whose cost is the resulting rule’s RE on the development set, with the conventional configuration seeded. Table 3 lists the parameter space and the winning configuration. The elite configurations agree on a higher selection pressure than the default (tournament size 7 versus 4) and a crossover probability close to the default (0.72–0.77); the tree-size cap does not discriminate (surviving elites span caps of 20, 30 and 40). Throughout the race the statistical tests discard configurations only weakly, itself evidence that performance is largely flat over the hyperparameter space.



Table 3: Hyperparameter space explored by irace (200 experiments; cost = development-set RE of the evolved rule) and resulting configurations.

| Parameter             | Range / values   | Default (seeded) | irace winner |
|-----------------------|------------------|------------------|--------------|
| Tournament size       | {2, 3, 4, 5, 7}  | 4                | 7            |
| Crossover prob. $p_c$ | [0.5, 0.95]      | 0.8              | 0.77         |
| Tree-size cap         | {20, 30, 40, 60} | 40               | 30           |
| Elitism               | {1, 2, 4}        | 2                | 2            |

We confirmed this on unseen instances following the pre-registered protocol applied to every design decision of this work: the winning configuration was re-evolved over 30 independent seeds and each resulting rule evaluated on all 70 instances. The tuned configuration brings *no* significant improvement— $18.99 \pm 1.33$  mean RE versus  $18.68 \pm 0.84$  for the default, with a paired Wilcoxon signed-rank test on the shared seeds far from significance ( $z = -0.96$ )—and if anything the stronger selection pressure increases seed-to-seed variance. Its one systematic effect is *structural*: at equal performance it evolves markedly smaller trees (26.4 versus 33.1 nodes on average), so the tuning budget buys readability rather than quality. The method is therefore robust to its evolutionary hyperparameters, and *every result reported in this paper was obtained with the default configuration* of Table 2.

## 6. Results

### 6.1. Evolution behaviour and seed stability

Over 30 independent evolutions, training RE converges to  $14.8 \pm 0.9$  (range 13.4–16.9), with most of the improvement realized by generation 25–30 (Figure 3). The evolved rules are structurally recognizable, combining the classical dispatching ingredients in interpretable ways. Figure 2 shows the

best rule of the campaign, obtained with seed 27: 37 nodes over 19 leaves. Writing  $A = NOR - PT + SLACK$  for the subtree that occurs twice in it, and flattening the nested minima, the rule is algebraically equivalent to:

$$\min\left(\frac{SLACK}{PT}, \max(ESTW, EST)\right) \cdot \min(A, WKR, NOR) + SLACK + \frac{1}{\min(A, WKR, SLACK) - EST} - \frac{WKR}{PT}, \quad (11)$$

which admits a reading in four parts. The last term is MWKR normalised by processing time, negated because the lowest priority is dispatched; the second penalises slack; the first weights slack per unit of processing time, capped by the worst-case start or by its uncertainty, by a measure of the remaining work; and the third is a correction that grows sharply when that measure approaches the earliest start. The evolved rule is thus a refinement of the most-work-remaining/earliest-start family that hand-crafted rules approximate only coarsely.

### 6.2. Generalization to the full benchmark

Table 4 reports single-rollout RE per size class over all 70 instances, as mean  $\pm$  standard deviation across the 30 evolved rules and for the single best rule. The best rule attains 17.3% global mean—evolved only on 20 $\times$ 15, its best class is the unseen 50 $\times$ 15 (13.3%), and the same pattern holds on average ( $13.5 \pm 1.4$ ). Because all terminals are per-operation attributes with no size-bound quantities, transfer requires no retraining or rescaling.

### 6.3. Position among constructive methods

Table 5 reports every constructive baseline under the protocol of this paper—one deterministic pass per instance, the same decoder and evaluator

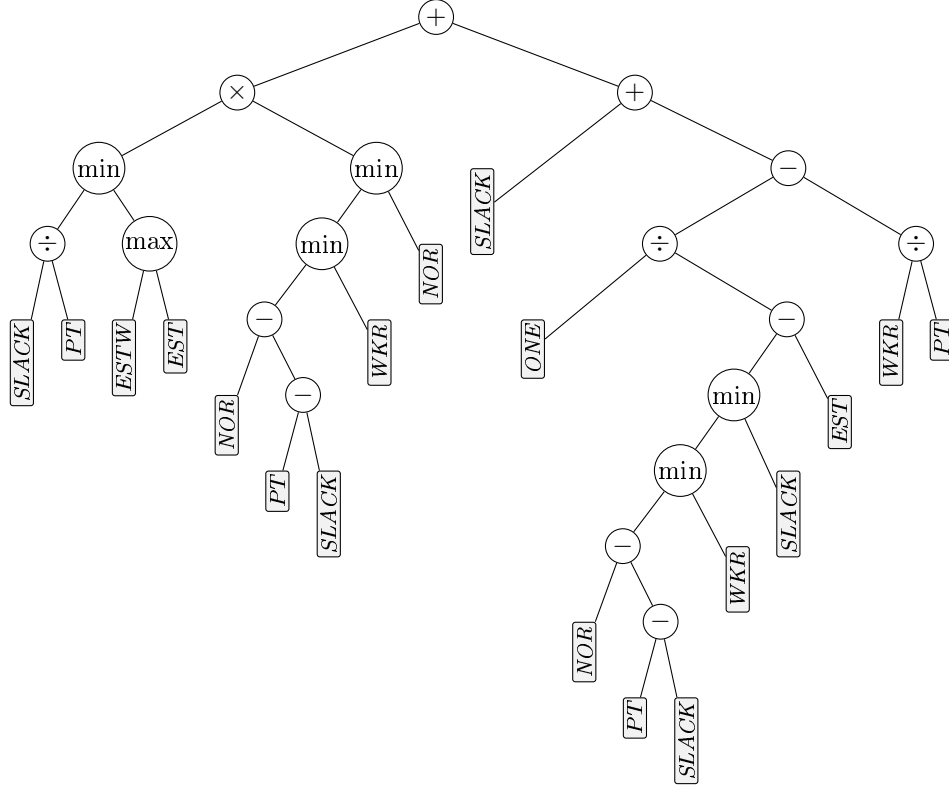


Figure 2: The best evolved rule of the campaign (seed 27), which attains 17.3% RE over the 70 instances. Circles are functions and shaded boxes the interval-aware terminals of Table 1; the operation with the lowest value of the expression is dispatched. The tree is shown exactly as evolved, without algebraic simplification.

for all methods—together with a uniformly random dispatcher as a reference point.

Two observations are needed to read the table correctly. First, the plain rules that ignore machine availability (SPT, LPT and the critical ratio CR, the latter being a due-date rule applied here without due dates) perform *worse than random* in this setting. This is a property of the schedule generation scheme, not a defect of the criteria: because a solution is de-

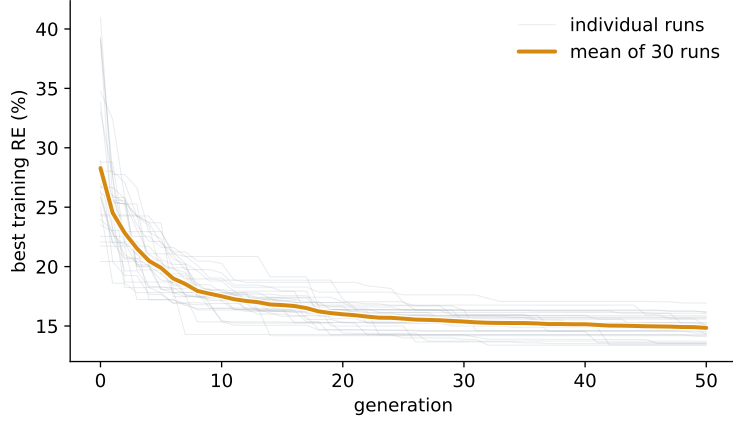


Figure 3: Best training fitness per generation over the 30 independent evolutions: each individual run and their mean.

Table 4: RE (%) per size class, single deterministic rollout, over all 70 Taillard interval instances: mean  $\pm$  sd across the 30 independently evolved rules, and the best evolved rule.

|            | 15 $\times$ 15 | 20 $\times$ 15 | 20 $\times$ 20 | 30 $\times$ 15 | 30 $\times$ 20 | 50 $\times$ 15 | 50 $\times$ 20 | All            |
|------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| Mean of 30 | 17.6 $\pm$ 1.2 | 17.5 $\pm$ 1.2 | 20.8 $\pm$ 1.4 | 21.2 $\pm$ 1.2 | 25.2 $\pm$ 1.2 | 13.5 $\pm$ 1.4 | 15.0 $\pm$ 1.2 | 18.7 $\pm$ 0.8 |
| Best rule  | 16.6           | 15.3           | 17.9           | 18.9           | 24.6           | <b>13.3</b>    | 14.4           | <b>17.3</b>    |

coded as a permutation whose operations are appended to their machines, systematically dispatching by processing time alone produces highly unbalanced permutations. The same criterion embedded in the Giffler–Thompson conflict set—which restores classical dispatching semantics—behaves as expected (SPT: 627% as a plain rule versus 70.6% under G&T). Second, the criteria that do account for the state of the shop (EST, MOR, MWKR) are comfortably better than random, and the strongest baseline overall is G&T-MWKR at 29.5%, which is the reference we use throughout.

Against that strongest baseline, the evolved rules average  $18.7 \pm 0.8$  RE over the 30 evolutions, and the best rule attains 17.3%.

Table 5: Constructive baselines on the 70 interval Taillard instances. All the rules are deterministic and build one schedule per instance, so the reported figures are the mean RE (%) over the 70 instances and the standard deviation *across instances*, which measures how uniform a method is over the benchmark rather than any run-to-run variability. The only stochastic entry is the uniformly random dispatcher, included as a reference and averaged over 10 independent dispatches per instance. Rules above the random line ignore machine availability and degrade below it under permutation-based decoding (see text).

| Method                   | RE (%)       | sd          |
|--------------------------|--------------|-------------|
| LPT                      | 703.6        | 168.0       |
| SPT                      | 627.2        | 152.2       |
| CR                       | 625.7        | 140.6       |
| <i>Random dispatcher</i> | <i>127.2</i> | <i>13.9</i> |
| G&T-SPT                  | 70.6         | 10.0        |
| MWKR                     | 64.7         | 10.4        |
| MOR                      | 45.5         | 10.1        |
| EST                      | 42.3         | 10.6        |
| G&T-MWKR                 | <b>29.5</b>  | 6.4         |
| GP rule (best of 30)     | <b>17.3</b>  | 5.3         |

The gap is not marginal and not an artefact of averaging: the best evolved rule is better than *both* baselines on every one of the 70 instances, and a paired Wilcoxon signed-rank test rejects equality against each baseline at  $p < 10^{-6}$  ( $z = -7.3$ ). Per-instance figures are given in Table A.10 (appendix). Crucially, this quality comes at no evaluation penalty over hand-crafted rules: evaluating the evolved expression tree over the eligible set costs the same, within measurement noise, as evaluating MOR or G&T-MWKR (the three differ by under 15% in per-schedule construction time in our implementation), since all reduce to one vectorized pass over eligible operations

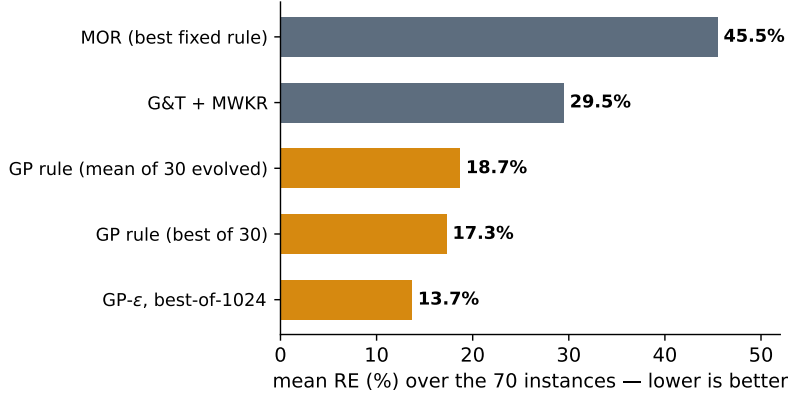


Figure 4: Constructive-method ladder on the 70 interval Taillard instances. Bars are single-solution methods except where noted.

per decision. Both cost and quality thus place the evolved rule squarely with the constructive methods, while closing much of the quality gap that separated them from search-based methods. Figure 4 summarizes the resulting ladder of constructive methods; the comparison against search-based metaheuristics is deferred to Section 6.4, where published results are available on a common benchmark.

#### 6.4. Cross-family generalization: the 12 classical instances

The 12 classical instances allow a direct confrontation with the *published* results of the IJSP metaheuristics, on their own benchmark and metric (Table 6). Recall that our rule was evolved on four interval Taillard  $20 \times 15$  instances and is deployed here *zero-shot*, on different instance families and sizes. The single deterministic pass averages 13.9% RE—far ahead of MOR (45.8%) and G&T-MWKR (30.1%) in the same computational class. More strikingly, the randomized extension GP- $\epsilon$  at best-of-1024 reaches 9.9%, matching the published average of the genetic algorithm of [1] (9.8%)—a per-instance meta-

Table 6: RE (%) on the 12 classical interval instances. Own methods (deterministic single pass and GP- $\varepsilon$  best-of- $N$ ; rule evolved on interval Taillard 20 $\times$ 15, applied zero-shot) versus published average results (30 runs) of the IJSP metaheuristics: GA [1] and the ABC variants [2, 3]. LB is the best-known lower bound of the expected makespan.

| Inst. | LB   | Own (zero-shot) |                        |                          | Published (avg of 30) |                   |       |       |
|-------|------|-----------------|------------------------|--------------------------|-----------------------|-------------------|-------|-------|
|       |      | GP              | GP- $\varepsilon_{64}$ | GP- $\varepsilon_{1024}$ | GA                    | ABC <sub>E3</sub> | fEABC | ESABC |
| FT10  | 930  | 8.1             | 7.0                    | 6.3                      | 5.2                   | 4.1               | 3.5   | 3.0   |
| FT20  | 1165 | 9.1             | 7.6                    | 6.2                      | 4.4                   | 1.7               | 1.8   | 1.8   |
| La21  | 1046 | 14.1            | 12.5                   | 9.6                      | 5.0                   | 5.0               | 4.2   | 4.0   |
| La24  | 935  | 11.1            | 8.7                    | 7.4                      | 6.3                   | 5.1               | 4.9   | 5.0   |
| La25  | 977  | 11.3            | 11.3                   | 6.6                      | 5.1                   | 3.9               | 3.4   | 2.7   |
| La27  | 1235 | 16.0            | 9.7                    | 9.2                      | 10.2                  | 4.7               | 4.6   | 4.1   |
| La29  | 1152 | 14.3            | 12.2                   | 11.4                     | 14.2                  | 8.6               | 7.4   | 7.0   |
| La38  | 1196 | 14.3            | 10.4                   | 8.0                      | 9.2                   | 6.9               | 6.1   | 5.8   |
| La40  | 1222 | 13.0            | 12.3                   | 8.1                      | 8.7                   | 4.2               | 4.6   | 4.1   |
| ABZ7  | 656  | 16.2            | 13.3                   | 11.5                     | 12.5                  | 7.3               | 6.6   | 6.7   |
| ABZ8  | 645  | 19.9            | 18.8                   | 16.1                     | 18.5                  | 12.1              | 11.1  | 10.9  |
| ABZ9  | 661  | 19.6            | 18.9                   | 18.8                     | 18.0                  | 13.1              | 11.6  | 11.2  |
| Mean  |      | 13.9            | 11.9                   | 9.9                      | 9.8                   | 6.4               | 5.8   | 5.5   |

heuristic evolving a population for every instance—and beating it on half of the instances. The purpose-built ABC variants and the tabu search remain clearly ahead (5.5–6.4%), as expected of iterative search: the point of this comparison is not to compete with them but to locate the constructive frontier—a single evolved rule, transferred across families without retraining, now covers a substantial part of the gap that used to separate dispatching rules from metaheuristics.

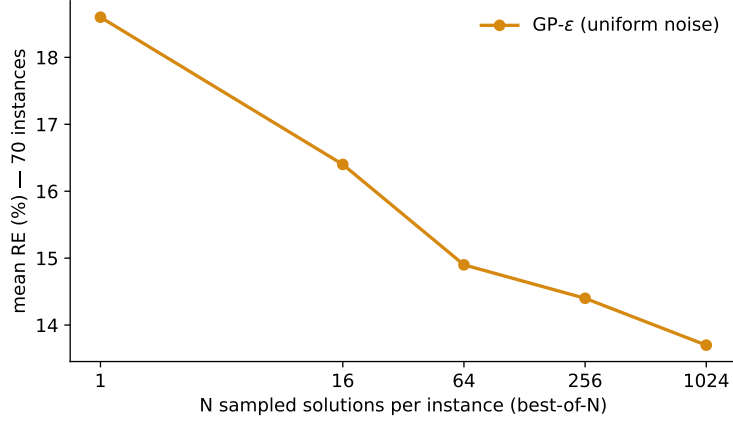


Figure 5: Best-of- $N$  curve of the evolved rule with uniform  $\varepsilon$ -randomization ( $\varepsilon=0.1$ ): mean RE over the 70 instances as the sampling budget doubles.

### 6.5. Sampling behaviour ( $GP-\varepsilon$ )

Best-of- $N$  over all 70 instances (pools of 1024  $\varepsilon$ -randomized samples per instance): 18.6% ( $N=1$ , deterministic), 16.4% (16), 14.9% (64), 14.4% (256), 13.7% (1024)—about  $-4.9$  points over ten doublings (Figure 5)—closing a substantial part of the gap to population-based methods while remaining a set of independent constructive passes: no solution is ever re-evaluated or refined, and the samples are trivially parallel.

## 7. Analysis

Beyond aggregate quality, this section examines *why* the evolved rules work: what they are made of, whether the interval-specific ingredients actually contribute, how robust the resulting schedules are under executional uncertainty, and whether they survive re-evaluation under other uncertainty models.



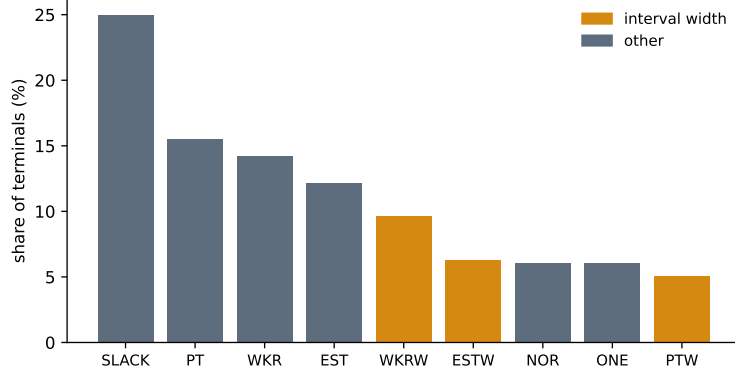


Figure 6: Terminal usage across the 30 evolved rules, as a share of all terminals. Interval-width terminals (amber) expose the magnitude of the uncertainty to the rule.

### 7.1. Rule anatomy and interpretability

A practical appeal of GP over neural policies is that its output is a readable expression. Evolved rules are compact—mean tree size 33 nodes (19–40), mean depth 10—so a rule can be inspected and simplified by hand. Figure 6 reports how often each terminal is used across the 30 evolved rules. The backbone is the classical scheduling triad—slack (*SLACK*), work remaining (*WKR*) and earliest start (*EST*)—together with worst-case processing time (*PT*), confirming that the evolution rediscovers and refines known dispatching logic rather than exploiting benchmark artefacts. The *interval-width* terminals (*PTW*, *ESTW*, *WKRW*) account for  $\approx 21\%$  of all terminal usage and appear in 28 of the 30 evolved rules: the evolution consistently reaches for uncertainty information, even though worst-case bounds carry the larger share of the signal. Whether that width information translates into better schedules is quantified by the ablation of Section 7.2 and the robustness analysis of Section 7.3.

## 7.2. The value of interval-awareness: terminal ablation

The interval-width terminals are the distinctive ingredient of this setting, so we isolate their contribution by ablation: we re-run the full evolution—30 independent seeds per configuration—with and without the width terminals (*PTW*, *ESTW*, *WKRW*) in the terminal set, everything else held fixed, and compare the resulting rules with a paired Wilcoxon signed-rank test on the shared seeds. Table 7 collects the outcome.

**Under the makespan objective, width information is not used productively.** Removing the three width terminals leaves performance unchanged:  $18.39 \pm 0.59$  mean RE over the 70 instances without them versus  $18.68 \pm 0.84$  with them, a difference that is not significant ( $z = 1.26$ ) and, if anything, favours the smaller terminal set. The same holds for the resulting schedules: the relative width of their predicted makespan interval is identical to two decimals ( $12.38 \pm 0.31$  with the width terminals versus  $12.38 \pm 0.22$  without them), and their  $\bar{\epsilon}$  robustness is indistinguishable (5.71 versus  $5.73 \times 10^{-3}$  for the best rule of each arm). So although the evolution does reach for the width terminals—they appear in 28 of the 30 rules (Section 7.1)—ablating them costs nothing: for minimising the expected makespan, the *position* of the intervals (their worst-case bounds) carries essentially all the exploitable signal.

**Under an uncertainty-aware objective, width information does pay.** This negative result is, on reflection, unsurprising: the fitness being optimised rewards only the makespan, so there is no incentive for a rule to act on the magnitude of the uncertainty. We therefore repeat the ablation with a *robust* fitness that explicitly rewards predictability, minimising  $\bar{C}_{\max} + \lambda(\bar{C}_{\max} - \underline{C}_{\max})$  with  $\lambda = 1$ : low worst case *and* narrow interval. Under this objective the ablation reverses. Rules evolved with the width ter-

minals produce schedules whose makespan interval is significantly narrower than those evolved without them ( $12.01 \pm 0.74$  versus  $12.42 \pm 0.21$  relative width; paired Wilcoxon  $z = 2.73$ ,  $p < 0.01$ ). The asymmetry is informative: deprived of width terminals, the evolution cannot narrow the interval below  $\approx 12.4\%$  *whatever* objective it is given, whereas with them it reaches 12.0% as soon as narrowness is rewarded. The gain is paid for in expected makespan (19.64 versus 18.70 RE), which is precisely the trade-off the robust objective asks for.

The weight  $\lambda$  governs that trade-off, and sweeping it traces the frontier between quality and predictability (Figure 7). Both quantities move monotonically with  $\lambda$ : as the weight grows from 0.5 to 4, the relative width of the makespan interval falls from  $12.16 \pm 0.51$  to  $10.30 \pm 1.43$  while the expected makespan degrades from  $19.09 \pm 1.45$  to  $23.04 \pm 3.26$  RE. At the far end of the sweep the schedules are therefore about two points narrower than anything the ablated rules can produce, at a cost of some four points of RE. The practitioner can choose an operating point along this curve; the rules deprived of width terminals have no such curve available to them, being pinned at  $\approx 12.4\%$  regardless of  $\lambda$ .

Together, these experiments characterise *when* interval awareness matters, rather than asserting that it always does: the width of an interval is actionable information for a dispatching rule, but only when the objective values predictability. A practitioner minimising expected makespan can safely use the worst-case attributes alone; one who needs schedules whose realised makespan is tightly predictable gains from exposing the widths, and can dial the balance through  $\lambda$ .

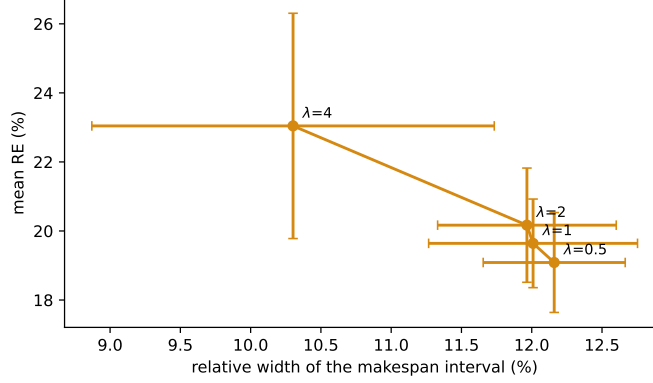


Figure 7: Quality–predictability frontier traced by the weight  $\lambda$  of the robust fitness (rules evolved *with* the interval-width terminals; mean  $\pm$  sd over independent evolutions, evaluated on the 70 instances). Increasing  $\lambda$  buys narrower makespan intervals at the price of a larger expected makespan.

### 7.3. Robustness under executional uncertainty

Relative error measures a-priori quality, but for the IJSP a schedule is ultimately *executed* under one realization of the durations. A good predictive schedule is one whose executed makespan stays close to its a-priori prediction. We quantify this with the  $\bar{\varepsilon}$  robustness of [1]: for a schedule with fixed processing order and predicted expected makespan  $E[C_{\max}]$  (interval midpoint), we draw  $K=1000$  realizations of the durations (uniform on each operation’s interval), execute the fixed order, and average the relative deviation of the executed makespan  $C_{\max}^{ex}$  from the prediction,  $\bar{\varepsilon} = \frac{1}{K} \sum_k |C_{\max}^{ex,k} - E[C_{\max}]| / E[C_{\max}]$ . Lower is more robust. To probe behaviour under growing uncertainty we also widen every interval symmetrically about its midpoint by +20% and +40% (clipping negative bounds to zero), re-evaluating the same processing orders.

The evolved rule does not merely produce lower-makespan schedules—

Table 7: Terminal ablation under two fitness objectives (30 independent evolutions per arm; rank-based tests between arms). *Width* is the relative width of the predicted makespan interval. Interval-width terminals do not help when the objective is the expected makespan—the two arms are then indistinguishable on both measures—but do produce significantly narrower schedules when the objective rewards predictability.

| Fitness objective                               | Terminal set   | RE (%)                     | Width (%)                          |
|-------------------------------------------------|----------------|----------------------------|------------------------------------|
| Makespan $E[\mathbf{C}_{\max}]$                 | full           | $18.68 \pm 0.84$           | $12.38 \pm 0.31$                   |
|                                                 | without widths | $18.39 \pm 0.59$           | $12.38 \pm 0.22$                   |
|                                                 | <i>test</i>    | <i>n.s.</i> ( $z = 1.26$ ) | <i>n.s.</i> ( $z = -0.12$ )        |
| Robust $\bar{C}_{\max} + \lambda \text{ width}$ | full           | $19.64 \pm 1.28$           | <b><math>12.01 \pm 0.74</math></b> |
|                                                 | without widths | $18.70 \pm 0.89$           | $12.42 \pm 0.21$                   |
|                                                 | <i>test</i>    | —                          | $z = 2.73, p < 0.01$               |

it produces *more robust* ones, on two complementary measures (Table 8). First, its predicted makespan interval is the *tightest*: a relative width of 12.6% against 13.8% (G&T-MWKR) and 14.4% (MOR), so the a-priori uncertainty attached to the schedule is smaller. Second, and more tellingly, its executed makespan tracks the prediction most closely:  $\bar{\varepsilon}$  is lower than for both baselines at the nominal width (5.8 versus 6.5 for MOR and 6.3 for G&T-MWKR), and its advantage *widens* as uncertainty grows (at +40%: 7.7 versus 8.9 and 8.6). Figure 8 shows the per-instance  $\bar{\varepsilon}$  distributions. It is worth being precise about the source of this advantage: as the ablation of Section 7.2 shows, it does *not* come from the rule reading interval widths—rules evolved without the width terminals are just as robust. It comes from producing better schedules in the first place: compact schedules with fewer and shorter critical paths accumulate less uncertainty along them, so both the predicted interval and the realised deviation shrink together.

Table 8: Robustness over the 70 instances (lower is better on both measures). *Rel. width*: relative width of the predicted makespan interval,  $(\overline{C}_{\max} - \underline{C}_{\max})/E[C_{\max}]$  (%).  $\bar{\varepsilon} (\times 10^3)$ : mean deviation of the executed makespan from its prediction, at nominal interval widths and widened by +20% and +40% (Monte Carlo,  $K=1000$  realizations, common random numbers across methods).

| Method   | Rel. width (%) | $\bar{\varepsilon} (\times 10^3)$ |            |            |
|----------|----------------|-----------------------------------|------------|------------|
|          |                | +0%                               | +20%       | +40%       |
| GP rule  | <b>12.6</b>    | <b>5.8</b>                        | <b>6.8</b> | <b>7.7</b> |
| G&T-MWKR | 13.8           | 6.3                               | 7.5        | 8.6        |
| MOR      | 14.4           | 6.5                               | 7.7        | 8.9        |

[TODO: tras la campaña: rehacer con la mejor regla de 30 semillas y anadir el brazo no-width (robustez interval-aware vs ablacion); test estadístico.]

#### 7.4. Transfer to other uncertainty models

The interval representation is one of several coexisting formalisms for duration uncertainty; deterministic (crisp) and fuzzy models are equally established [27]. A natural concern is whether schedules built by a rule evolved *for intervals* remain good when the durations are ultimately handled under a different model. We test this by re-decoding the same processing orders under two alternative arithmetics, without any re-evolution: (i) *crisp*, executing each sequence with the original Taillard durations (the midpoint scenario of each interval instance); and (ii) *triangular fuzzy* (TFN), where each duration becomes the triangle  $(\underline{p}, p, \bar{p})$  and schedules are built with fuzzy arithmetic and ranked by the expected value  $E[\tilde{C}] = (a + 2b + c)/4$ . The evaluation uses the GP- $\varepsilon$  pools of Section 6 (1024 sequences per instance, all 70 instances), so both quality and *ranking* transfer can be measured.

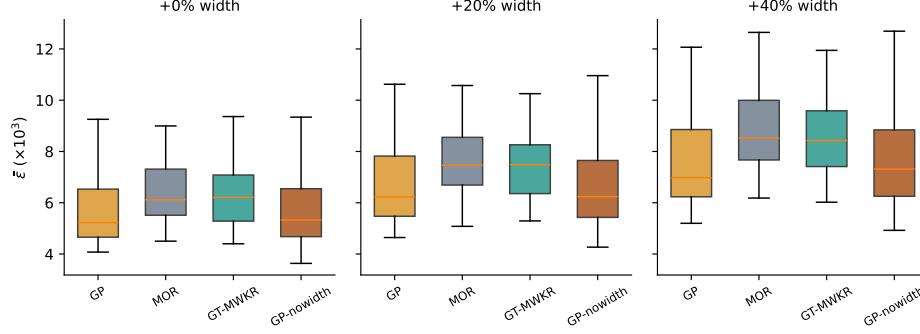


Figure 8: Per-instance  $\bar{\epsilon}$  robustness by method at each uncertainty level (lower is better). The evolved rule’s robustness advantage over the constructive baselines grows with interval width.

Transfer is essentially lossless (Table 9). The best sequence per pool scores 13.0% RE under crisp and 13.2% under TFN evaluation, in line with its interval quality; pool means are unchanged (21.0–21.2 versus 21.4); and the interval ranking of each pool is preserved almost exactly under both models (mean Spearman correlation 0.994 crisp, 0.998 fuzzy)—the best solution under interval arithmetic is, for practical purposes, also the best crisp and fuzzy solution. Notably, there is no “robustness tax”: evaluating crisp, the interval-selected sequences are marginally *better* than their interval RE suggests. For a practitioner this means the choice of uncertainty formalism can be deferred: schedules produced by the interval-evolved rule survive re-evaluation under the other standard models of the literature.

### 7.5. Limitations

Three limitations bound the scope of our conclusions. First, all rules operate inside a semi-active SGS; active (insertion-based) generation schemes are known to help weaker constructors, and although our evolved rules pro-

Table 9: Transfer of the GP- $\varepsilon$  pools (1024 sequences per instance, 70 instances) to other uncertainty models: the same processing orders re-decoded under each arithmetic. Spearman: mean per-instance rank correlation between the interval ranking and the re-decoded ranking.

| Evaluation model                 | Best of pool (%) | Pool mean (%) | Spearman |
|----------------------------------|------------------|---------------|----------|
| Interval (native)                | 13.4             | 21.4          | —        |
| Crisp (midpoint scenario)        | 13.0             | 21.0          | 0.994    |
| Triangular fuzzy, $E[\tilde{C}]$ | 13.2             | 21.2          | 0.998    |

duce near-active schedules already, co-evolving rules with an active SGS in the loop remains untested. Second, the uncertainty model is the  $\pm 15\%$  symmetric interval scheme standard in the IJSP literature; other widths and asymmetric intervals may change the relative value of uncertainty information. Third, our runtime figures are relative (evolved rules cost the same as hand-crafted ones per pass); absolute times depend on the research-grade Python implementation and would shrink by orders of magnitude in an optimized production implementation, without affecting any ranking reported here.

## 8. Conclusions and Future Work

We presented what is, to the best of our knowledge, the first GP hyper-heuristic for job shop scheduling with interval durations, built on a terminal set that exposes interval widths as first-class attributes. The empirical conclusions are three. (i) Evolved rules dominate every deterministic constructive baseline on the 70-instance benchmark by a wide, statistically significant margin ( $18.7 \pm 0.8$  over 30 evolutions, 17.3% for the best rule, versus 29.5% for the strongest baseline—which the best rule beats on all 70 instances) at



the cost of a single constructive pass, and (ii) they generalize zero-shot across all size classes—evolved on  $20 \times 15$ , best on the unseen  $50 \times 15$ —because the representation contains no size-bound quantities. (iii) The method is robust to its evolutionary hyperparameters: a 200-experiment irace study confirmed over 30 evolutions per configuration yields no significant gain over conventional settings, only systematically smaller trees at equal performance. Beyond aggregate quality, the analysis pins down *when* the interval-specific ingredients matter: exposing the widths of the uncertain quantities does not improve the expected makespan—the worst-case bounds already carry that signal—but it does let the evolution produce significantly narrower makespan intervals once the objective rewards predictability. Interval awareness is thus not a free improvement but a lever, available when the application values predictable schedules over nominally shorter ones.

Future work follows three lines. First, richer symbolic models: ensembles of complementary rules [22] and co-evolution of construction and tie-breaking. Second, using evolved rules as cheap, high-quality population initializers for the metaheuristics that lead the IJSP: the pools of diverse, high-quality sequences produced by GP- $\varepsilon$  are directly usable as seeded initial populations, and quantifying the effect of such seeding is ongoing work. Third, a controlled comparison with learned neural constructive policies under matched training and inference budgets—the head-to-head evaluation that the recent literature [21] identifies as missing.

### **CRedit authorship contribution statement**

**Hernán Díaz:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original

draft, Writing – review & editing, Visualization, Project administration.

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## Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

The GP hyper-heuristic is implemented from scratch in pure Python/NumPy, with no evolutionary-computation dependencies; evolved rules are stored as plain JSON expression trees and evaluated as drop-in dispatching strategies by the same environment used by every baseline. The benchmark instances, the generated solutions and rules, and the analysis scripts are shared in a Zenodo repository. [\[TODO: DOI de Zenodo al depositar\]](#)

## Appendix A. Per-instance results

Table A.10 lists, for each of the 70 interval Taillard instances, the crisp lower bound and the single-pass RE (%) of the three deterministic constructive methods.

[TODO: tabla por instancia: se refresca con la mejor regla de la campana 30x3]

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Table A.10: Per-instance RE (%) of the constructive methods (single deterministic pass).

LB is the crisp Taillard lower bound.

| Inst. | LB   | MOR  | G&T  | GP   | Inst. | LB   | MOR  | G&T  | GP   |
|-------|------|------|------|------|-------|------|------|------|------|
| TA1   | 1231 | 29.7 | 26.1 | 18.6 | TA36  | 1819 | 57.4 | 42.8 | 20.0 |
| TA2   | 1244 | 34.8 | 27.8 | 15.5 | TA37  | 1771 | 43.0 | 35.6 | 23.4 |
| TA3   | 1218 | 42.9 | 33.2 | 16.1 | TA38  | 1673 | 50.6 | 28.0 | 21.1 |
| TA4   | 1175 | 62.6 | 27.7 | 17.0 | TA39  | 1795 | 50.2 | 28.9 | 18.7 |
| TA5   | 1224 | 50.7 | 33.0 | 8.8  | TA40  | 1651 | 42.7 | 42.3 | 21.5 |
| TA6   | 1238 | 36.6 | 18.2 | 16.7 | TA41  | 1906 | 53.5 | 45.0 | 32.3 |
| TA7   | 1227 | 39.4 | 26.1 | 21.8 | TA42  | 1884 | 61.3 | 38.1 | 23.0 |
| TA8   | 1217 | 41.3 | 25.0 | 13.9 | TA43  | 1809 | 61.4 | 39.8 | 24.2 |
| TA9   | 1274 | 38.2 | 28.2 | 17.9 | TA44  | 1948 | 58.8 | 34.7 | 24.4 |
| TA10  | 1241 | 38.0 | 22.1 | 19.6 | TA45  | 1997 | 59.5 | 33.7 | 16.2 |
| TA11  | 1357 | 49.9 | 30.7 | 16.5 | TA46  | 1957 | 59.7 | 31.6 | 24.5 |
| TA12  | 1367 | 47.8 | 34.6 | 15.4 | TA47  | 1807 | 78.3 | 30.0 | 23.2 |
| TA13  | 1342 | 46.5 | 26.9 | 14.9 | TA48  | 1912 | 52.2 | 33.7 | 26.3 |
| TA14  | 1345 | 44.1 | 29.7 | 9.7  | TA49  | 1931 | 46.9 | 35.3 | 22.8 |
| TA15  | 1339 | 51.3 | 30.6 | 16.4 | TA50  | 1833 | 56.2 | 46.2 | 29.7 |
| TA16  | 1360 | 35.8 | 34.0 | 18.5 | TA51  | 2760 | 38.2 | 34.7 | 26.3 |
| TA17  | 1462 | 50.1 | 17.7 | 12.8 | TA52  | 2756 | 38.1 | 26.5 | 10.5 |
| TA18  | 1377 | 54.2 | 31.8 | 18.0 | TA53  | 2717 | 28.2 | 18.3 | 11.0 |
| TA19  | 1332 | 43.2 | 29.3 | 13.7 | TA54  | 2839 | 23.6 | 16.1 | 4.7  |
| TA20  | 1348 | 43.7 | 24.1 | 16.9 | TA55  | 2679 | 37.6 | 29.2 | 13.9 |
| TA21  | 1642 | 40.8 | 28.0 | 15.4 | TA56  | 2781 | 44.8 | 19.8 | 10.9 |
| TA22  | 1561 | 46.3 | 30.8 | 24.6 | TA57  | 2943 | 29.9 | 20.0 | 12.7 |
| TA23  | 1518 | 45.2 | 35.3 | 22.5 | TA58  | 2885 | 30.5 | 26.3 | 13.7 |
| TA24  | 1644 | 36.2 | 23.8 | 14.4 | TA59  | 2655 | 45.2 | 28.2 | 19.0 |
| TA25  | 1558 | 53.2 | 33.8 | 17.4 | TA60  | 2723 | 25.3 | 19.8 | 10.1 |
| TA26  | 1591 | 52.4 | 31.2 | 18.0 | TA61  | 2868 | 48.7 | 25.4 | 17.2 |
| TA27  | 1652 | 49.8 | 31.1 | 16.9 | TA62  | 2869 | 43.5 | 27.6 | 14.9 |
| TA28  | 1603 | 38.8 | 26.8 | 17.7 | TA63  | 2755 | 46.0 | 23.8 | 11.8 |
| TA29  | 1583 | 56.4 | 32.0 | 12.3 | TA64  | 2702 | 43.3 | 28.4 | 11.5 |
| TA30  | 1528 | 53.3 | 36.1 | 19.8 | TA65  | 2725 | 44.7 | 30.0 | 19.7 |
| TA31  | 1764 | 34.4 | 34.4 | 15.2 | TA66  | 2845 | 38.1 | 28.0 | 14.8 |
| TA32  | 1774 | 55.7 | 33.3 | 21.2 | TA67  | 2825 | 58.4 | 26.1 | 12.7 |
| TA33  | 1788 | 41.6 | 34.8 | 22.7 | TA68  | 2784 | 45.8 | 19.1 | 10.3 |
| TA34  | 1828 | 34.8 | 31.8 | 16.4 | TA69  | 3071 | 38.0 | 22.8 | 11.4 |
| TA35  | 2007 | 30.5 | 24.1 | 9.2  | TA70  | 2995 | 52.3 | 23.4 | 20.2 |